GSEM, University of Geneva
Forecasting with Applications in Business ('S411031')
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Practical 3
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Time Series Decomposition: Additive and Multiplicative Models

Exercise 1. Additive Decomposition: Beer Production Data

- Load the dataset `aus_production` from the package `tsibbledata`, read the dataset's description and extract quarterly beer production from 1992 Q1 to 2000 Q4.
- Plot the time series.
- Perform a manual decomposition using LOESS for trend and average detrended values per quarter for seasonality.
- Perform an STL decomposition using `STL()` from the `fpp3` package.
- Plot and compare the trend, seasonal, and remainder components from the manual and STL decompositions.

Exercise 2. Synthetic Additive Time Series

- Generate a quarterly additive series with linear trend, seasonal component, and Gaussian noise:

$$Y_t = T_t + S_t + \varepsilon_t$$

- Plot the generated series.
- Perform a manual decomposition using LOESS for the trend and average detrended values per quarter for seasonality.
- Perform STL decomposition.
- Compare the true value of the trend, seasonal and error components with their estimated values using a manually implemented STL decomposition and the implemented STL method.

Exercise 3. Multiplicative Decomposition: AirPassengers

- Use the dataset `AirPassengers` from the package `datasets`. Plot the series.
- Transform the series using logarithm to perform a multiplicative decomposition.
- Perform a manual decomposition (LOESS for trend, average detrended by month for seasonality) and STL decomposition.
- Plot and compare the trend, seasonal, and remainder components from the manual and STL decompositions on log-scale.

Exercise 4. Synthetic Multiplicative Time Series

- Generate a synthetic monthly multiplicative series:

$$Y_t = T_t \times S_t \times \varepsilon_t$$

  with increasing trend, multiplicative seasonal pattern, and multiplicative noise.

- Plot the series.
- Transform using logarithm and perform manual decomposition (trend via LOESS, seasonal via average detrended per month) and STL decomposition.
- Compare the true value of the trend, seasonal and error components with their estimated values using a manually implemented STL decomposition and the implemented STL method. (all on log-scale).